# 1. Key Constraints

## Region

$$\sigma_{R1.region_id=R2.region_id \wedge (R1.region_name \neq R2.region_name \vee R1.country \neq R2.country)} (\rho_{R1}(Region) \times \rho_{R2}(Region)) = \emptyset$$

## GPUType

All attributes make up the key

## SpecificGPUType

$$\sigma_{S1.gpu_type_name = S2.gpu_type_name \wedge S1.memory_gb \neq S2.memory_gb} (\rho_{S1}(SpecificGPUType) \times \rho_{S2}(SpecificGPUType)) = \emptyset$$

## AIModel

$$\sigma_{A1.model_id = A2.model_id \wedge A1.model_name \neq A2.model_name} (\rho_{A1}(AIModel) \times \rho_{A2}(AIModel)) = \emptyset$$

## GPUServer

$$\sigma_{G1.machine_id = G2.machine_id \wedge (G1.cpu_capacity \neq G2.cpu_capacity \vee G1.gpu_capacity \neq G2.gpu_capacity \vee G1.region_id \neq G2.region_id \vee G1.gpu_type_name \neq G2.gpu_type_name)} (\rho_{G1}(GPUServer) \times \rho_{G2}(GPUServer)) = \emptyset$$

## InferenceRequest

$$\sigma_{I1.request_id = I2.request_id \wedge (I1.submitted_at \neq I2.submitted_at \vee I1.input_size \neq I2.input_size \vee I1.status \neq I2.status \vee I1.response_time_ms \neq I2.response_time_ms \vee I1.region_id \neq I2.region_id \vee I1.model_id \neq I2.model_id \vee I1.machine_id \neq I2.machine_id)} (\rho_{I1}(InferenceRequest) \times \rho_{I2}(InferenceRequest)) = \emptyset$$

## MachineMetric

$$\sigma_{M1.machine_id = M2.machine_id \wedge M1.metric_id = M2.metric_id \wedge (M1.start_time \neq M2.start_time \vee M1.end_time \neq M2.end_time \vee M1.worker_load \neq M2.worker_load \vee M1.gpu_utilization \neq M2.gpu_utilization \vee M2.gpu_utilization \neq M2.cpu_utilization)} (\rho_{M1}(MachineMetric) \times \rho_{M2}(MachineMetric)) = \emptyset$$

Hosts

All attributes make up the key

CompatibleWith

All attributes make up the key

2. Referential Integrity

$\Pi_{gpu_type_name}(SpecificGPUType) \subseteq \Pi_{gpu_type_name}(GPUType)$

$\Pi_{region_id}(GPUServer) \subseteq \Pi_{region_id}(Region)$

$\Pi_{gpu_type_name}(GPUServer) \subseteq \Pi_{gpu_type_name}(GPUType)$

$\Pi_{region_id}(InferenceRequest) \subseteq \Pi_{region_id}(Region)$

$\Pi_{model_id}(InferenceRequest) \subseteq \Pi_{model_id}(AIModel)$

$\Pi_{machine_id}(InferenceRequest) \subseteq \Pi_{machine_id}(GPUServer)$

$\Pi_{machine_id}(MachineMetric) \subseteq \Pi_{machine_id}(GPUServer)$

$\Pi_{model_id}(Hosts) \subseteq \Pi_{model_id}(AIModel)$

$\Pi_{machine_id}(Hosts) \subseteq \Pi_{machine_id}(GPUServer)$

$\Pi_{model_id}(CompatibleWith) \subseteq \Pi_{model_id}(AIModel)$

$\Pi_{gpu_type_name}(CompatibleWith) \subseteq \Pi_{gpu_type_name}(GPUType)$

3. Domain & Tuple Constraints

$\sigma_{cpu_capacity \leq 0 \vee memory_capacity \leq 0 \vee gpu_capacity \leq 0}(GPUServer) = \emptyset$

$\sigma_{memory_gb \leq 0}(SpecificGPUType) = \emptyset$

$\sigma_{end_time < start_time}(MachineMetric) = \emptyset$

$\sigma_{worker_load < 0 \vee gpu_utilization < 0 \vee cpu_utilization < 0}(MachineMetric) = \emptyset$

$\sigma_{response_time_ms < 0}(InferenceRequest) = \emptyset$

$\sigma_{input_size \leq 0}(InferenceRequest) = \emptyset$

$\sigma_{status \neq "Failed" \wedge status \neq "Running" \wedge status \neq "Terminated"}(InferenceRequest) = \emptyset$

4. Cross-Table Constraints

- A server must only host AI models compatible with its GPU type:

$$\pi_{model_id,\ gpu_type_name}(Hosts \bowtie GPUserver) \setminus CompatibleWith = \varnothing$$

- An inference request can only run on a server that hosts the requested model:

$$\pi_{model_id,\ machine_id}(InferenceRequest) \subseteq Hosts$$